

[This question paper contains 16 printed pages.]

Your Roll No.....

Sr. No. of Question Paper : 4432 G

Unique Paper Code : 32347507

Name of the Paper : Data Analysis and  
Visualization

Name of the Course : B.Sc. (Hons.) Computer  
Science

Semester : V

Duration : 3 Hours Maximum Marks : 75

### Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Question No. 1 is compulsory.
3. Attempt any **four** questions out of Q.2 to Q.7.
4. Parts of a question must be answered together.

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1. (a) Provide code to create a time-series with two index labels- 2011/9/01 and 2011/9/02. Assign random values. (2)

(b) What will be the output of the following codes?

(i) (2)

```
import numpy as np
arr = np.array([[1,2,3,4,5],[6,7,8,9,10]])
print(arr[1,-1], arr[-1:])
```

(ii) (2)

```
List = [str[::-1] for str in ('happy','go','lucky')]
print(List)
```

(c) Reshape the following array to dimension (2,6)

[[3,4,5,6], [7,8,9,10], [11,12,13,14]] (2)

(d) Python is a strongly “typed” language. Comment.

(2)

- (c) Give the output of the following code and identify the role of *is\_unique* attribute in the code. (2)

```
import pandas as pd
series = pd.Series([4,5,1,2,3,3,4,5,6])
print(series)
print("Is Unique: ",series.is_unique)
```

- (f) Differentiate between mutable and immutable objects. (2)
- (g) Write a program to create the given dataframe. (3)

|   | id | value |
|---|----|-------|
| 0 | 1  | a     |
| 1 | 1  | a     |
| 2 | 2  | b     |
| 3 | 3  | None  |
| 4 | 3  | a     |
| 5 | 4  | a     |
| 6 | 4  | None  |
| 7 | 4  | b     |

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Further, split it into groups and count unique values of 'value' column.

- (h) Provide the output of the following code: (3)

```
from datetime import datetime, date, time
dt = datetime(2011, 10, 29, 20, 30, 21)
dt2 = datetime(2011, 11, 15, 22, 30)
delta = dt2 - dt

print(delta)
print(type(delta))
print(dt.replace(minute=0, second=0))
```

- (i) Consider the given dataframe *df* containing data of students admitted in the college. (3)

| Id | Name | Age | Section | City    | Gender | Marks |
|----|------|-----|---------|---------|--------|-------|
| S0 | Anit | 10  | A       | Gurgaon | M      | 60    |
| S1 | Alka | 22  | B       | Delhi   | F      | 80    |
| S2 | Sid  | 13  | C       | Mumbai  | M      | 60    |
| S3 | Ruhi | 21  | B       | Delhi   | F      | 55    |
| S4 | Nehu | 12  | B       | Mumbai  | F      | 60    |
| S5 | Geet | 11  | A       | Delhi   | F      | 56    |
| S6 | Om   | 17  | A       | Mumbai  | M      | 45    |

Set the first column '*Id*' as the row index of the given dataframe *df*. Create a pivot table of *df* to display the total number of admissions as per '*Section*' and '*Gender*'.

- (j) (i) Provide the output of the following code :  
(4)

```
df = pd.DataFrame({
    'a':np.arange(1,7),
    'b':np.arange(7,13),
    'c':np.arange(12,18),
    'd':np.arange(17,23),
    'e':np.arange(23,29),
    'f':np.arange(29,35)},
    columns=['a', 'b', 'c', 'd', 'e', 'f'],
    index=['Svaksh', 'Sarah', 'Svaraj', 'Rivika',
    'Rahul', 'Geet'])
print(df)
df.iloc[2:4,[1,2]]=np.NaN
print(df)

mapping = {'a': 'red', 'b': 'red', 'c': 'blue', 'd':
'blue', 'e': 'red', 'f': 'orange'}
```

- (ii) Using the above dataframe, group *df* by mapping and find the sum.

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- (k) Consider the following dataset to perform the following operations :  
(4)

|   | Age | Section | City    | Gender | Favourite_color |
|---|-----|---------|---------|--------|-----------------|
| 0 | 10  | A       | Gurgaon | M      | red             |
| 1 | 22  | B       | Delhi   | F      | NaN             |
| 2 | 13  | C       | Mumbai  | F      | yellow          |
| 3 | 21  | B       | Delhi   | M      | NaN             |
| 4 | 12  | B       | Mumbai  | M      | black           |
| 5 | 11  | A       | Delhi   | M      | green           |
| 6 | 17  | A       | Mumbai  | F      | red             |

- (i) Find all the rows where Age is greater than or equal to 12 and the Gender is male.
- (ii) If Age is greater than 20, then use the loc function to update Section with "S" and City with Pune.
- (iii) Select rows 1 to 2 with columns 2 to 3

using iloc.

(L) What is the output of the following code : (4)

```
import pandas as pd
fruits=['apple','orange','apple','apple']*2
N=len(fruits)

print(N)
df=pd.DataFrame({'fruit':fruits,'basket_ID':
np.arange(N),'count':
np.random.randint(3,15,size=N),'weight':
np.random.uniform(0,4, size=N)})
print(df)
Fruit_cat=df['fruit'].astype('category')
print(Fruit_cat)
print(df.dtypes)
```

2. (a) Differentiate between :

(i) qcut and cut methods

(ii) Pandas.merge and pandas.concat (2)

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(b) Consider the following numeric grades (out of 4).  
Formulate bins for the given grades as per the  
following condition : (3)

|                     |           |
|---------------------|-----------|
| Below 2.5           | Very bad  |
| Between 2.5 to 3    | Bad       |
| Between 3 to 3.25   | Average   |
| Between 3.25 to 3.5 | Good      |
| Between 3.5 to 3.75 | Very good |
| Between 3.75 to 4   | Excellent |

(c) Given the following dataframe, provide the output  
for the following commands : (5)

|    | ord no  | purch amt | ord date   | customer_id |
|----|---------|-----------|------------|-------------|
| 0  | NaN     | NaN       | NaN        | NaN         |
| 1  | NaN     | 270.65    | 2012-09-10 | 3001.0      |
| 2  | 70002.0 | 65.26     | NaN        | 3001.0      |
| 3  | NaN     | NaN       | NaN        | NaN         |
| 4  | NaN     | 948.50    | 2012-09-10 | 3002.0      |
| 5  | 70005.0 | 2400.60   | 2012-07-27 | 3001.0      |
| 6  | NaN     | 5760.00   | 2012-09-10 | 3001.0      |
| 7  | 70010.0 | 1983.43   | 2012-10-10 | 3004.0      |
| 8  | 70003.0 | 2480.40   | 2012-10-10 | 3003.0      |
| 9  | 70012.0 | 250.45    | 2012-06-27 | 3002.0      |
| 10 | NaN     | 75.29     | 2012-08-17 | 3001.0      |
| 11 | NaN     | NaN       | NaN        | NaN         |

(i) `df.dropna(thresh=2)`

(ii) `df.dropna(how='all')`

(iii) `df.dropna(how='all', axis=1)`

(iv) `df.isnull()`

(v) `df.isnull().values.any()`

3. (a) Write the code to read each row of a given csv file. Skip the header of the file while reading. Also print the number of rows and the field names.

(6)

(b) (i) Differentiate between `ffill` and `bfill`. (4)

(ii) Provide the output of the given code :

```
import pandas as pd
obj3 = pd.Series(['blue', 'purple', 'yellow'],
                 index=[0, 2, 4])
print(obj3.reindex(range(6), method='ffill'))
print(obj3.reindex(range(6), method='bfill'))
```

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4. (a) Consider following dataframe.

| ord no  | purch amt | ord date   | customer id | salesman id |
|---------|-----------|------------|-------------|-------------|
| 70009.0 | 890.00    | 2012-09-11 | 3004.0      | 5001        |
| 70002.0 | 270.65    | 2012-09-10 | 3001.0      | 5006        |
| 70007.0 | 65.26     | 2012-09-11 | 3001.0      | 5005        |
| 70008.0 | 78.00     | 2012-09-10 | 3002.0      | 5003        |
| 70006.0 | 948.50    | 2012-09-17 | 3002.0      | 5002        |
| 70005.0 | 2400.60   | 2012-07-27 | 3001.0      | 5001        |
| 70004.0 | 5760.00   | 2012-09-10 | 3001.0      | 5003        |
| 70010.0 | 1983.43   | 2012-10-10 | 3004.0      | 5006        |
| 70003.0 | 2480.40   | 2012-10-10 | 3003.0      | 5005        |
| 70012.0 | 250.45    | 2012-06-27 | 3002.0      | 5002        |
| 70034.0 | 75.29     | 2012-08-17 | 3001.0      | 5004        |
| 70022.0 | 56.90     | 2012-06-27 | 3003.0      | 5005        |

With respect to the above dataframe, write the code for the following :

- (i) Group the data on the column `ord_date` and calculate the total purchase amount `purch_amt` year wise and month wise.

(2)

(ii) Group the data on the column *customer\_id* and create a list of order date *ord\_date* for each group. (2)

(iii) Group on the columns *customer\_id*, *salesman\_id* and then sort sum of *purch\_amt* within the groups. (2)

(b) (i) Write a generator function to print Fibonacci numbers. (4)

(ii) What is the output of the following code :

```
def simpleGeneratorFunc():
    yield 1
    yield 2
x = simpleGeneratorFunc()

print(next(x))
print(next(x))
```

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5. (a) Give output of the following code. Justify your answer. (2)

```
var=(1, 2, (3,4))
var[1]='geet'
print(var)
```

(b) Write the code to merge the two given datasets using key1, key2. (4)

data1:

|   | key1 | key2 | P  | Q  |
|---|------|------|----|----|
| 0 | K0   | K0   | P0 | Q0 |
| 1 | K0   | K1   | P1 | Q1 |
| 2 | K1   | K0   | P2 | Q2 |
| 3 | K2   | K1   | P3 | Q3 |

data2:

|   | key1 | key2 | R  | S  |
|---|------|------|----|----|
| 0 | K0   | K0   | R0 | S0 |
| 1 | K1   | K0   | R1 | S1 |
| 2 | K1   | K0   | R2 | S2 |
| 3 | K2   | K0   | R3 | S3 |

- (c) Write the code to split the given dataset into groups based on *customer\_id* and create a list of order date *ord\_date* for each group.

(4)

|    | ord no  | purch amt | ord date   | customer id |
|----|---------|-----------|------------|-------------|
| 0  | 70009.0 | 890.00    | 2012-09-11 | 3004.0      |
| 1  | 70002.0 | 270.65    | 2012-09-10 | 3001.0      |
| 2  | 70007.0 | 65.26     | 2012-09-11 | 3001.0      |
| 3  | 70008.0 | 78.00     | 2012-09-10 | 3002.0      |
| 4  | 70006.0 | 948.50    | 2012-09-17 | 3002.0      |
| 5  | 70005.0 | 2400.60   | 2012-07-27 | 3001.0      |
| 6  | 70004.0 | 5760.00   | 2012-09-10 | 3001.0      |
| 7  | 70010.0 | 1983.43   | 2012-10-10 | 3004.0      |
| 8  | 70003.0 | 2480.40   | 2012-10-10 | 3003.0      |
| 9  | 70012.0 | 250.45    | 2012-06-27 | 3002.0      |
| 10 | 70034.0 | 75.29     | 2012-08-17 | 3001.0      |
| 11 | 70022.0 | 56.90     | 2012-06-27 | 3003.0      |

6. (a) Create a Timeseries Dataframe with date range 01-02-2022 to 30-02-2022 with 1 min frequency interval. The dataframe has two columns populated with random values.

(3)

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- (b) Identify the need to resample Timeseries data.

(2)

- (c) Consider following dataset.

| Datetime               | value1 | value2 | value3 |
|------------------------|--------|--------|--------|
| 2020-01.01 00:00:00    | 2      | 92     | 56     |
| 2020-01.01 00:01:00    | 9      | 78     | 80     |
| 2020-01.01 00:02:00    | 69     | 83     | 43     |
| 2020-01.01 00:03:00    | 47     | 62     | 45     |
| 2020-01.01 00:04:00    | 47     | 90     | 13     |
| ...                    | ...    | ...    | ...    |
| 2020-02.27 23:56:00    | 73     | 81     | 35     |
| 2020-02.27 23:57:00    | 20     | 66     | 58     |
| 2020-02.27 23:58:00    | 42     | 16     | 48     |
| 2020-02.27 23:59:00    | 32     | 40     | 19     |
| 2020-02.28 00:00:00    | 37     | 63     | 95     |
|                        |        |        |        |
| 83521 rows x 3 columns |        |        |        |

- (i) Resample for 10min with sum function for *value1*, mean for *value2* and max for *value 3*.

(3)

- (ii) Downsample data to 30s. (2)

7. (a) Create a DataFrame of 8 rows and 8 columns containing random integers in the range of 1 to 10. Compute the correlation of each row with the preceding row. (2)

(b) Consider the following table that lists the last week Delhi's AQI.

| AQI | Date       |
|-----|------------|
| 67  | 2/10/2022  |
| 79  | 3/10/2022  |
| 80  | 4/10/2022  |
| 90  | 5/10/2022  |
| 99  | 6/10/2022  |
| 110 | 7/10/2022  |
| 112 | 8/10/2022  |
| 140 | 9/10/2022  |
| 165 | 10/10/2022 |
| 178 | 11/10/2022 |

- (i) Plot a line graph showing AQI (Air Quality Index) against date with line colour as red, line width as "4pixels" and dashed line style. (4)

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- (ii) Add title "Delhi AQI for last ten days". (1)

(iii) Set label for x-axis "Date" and y-axis "AQI". (1)

(iv) Show grids in the background. (1)

(v) Set marker as '\*'. (1)

(1000)